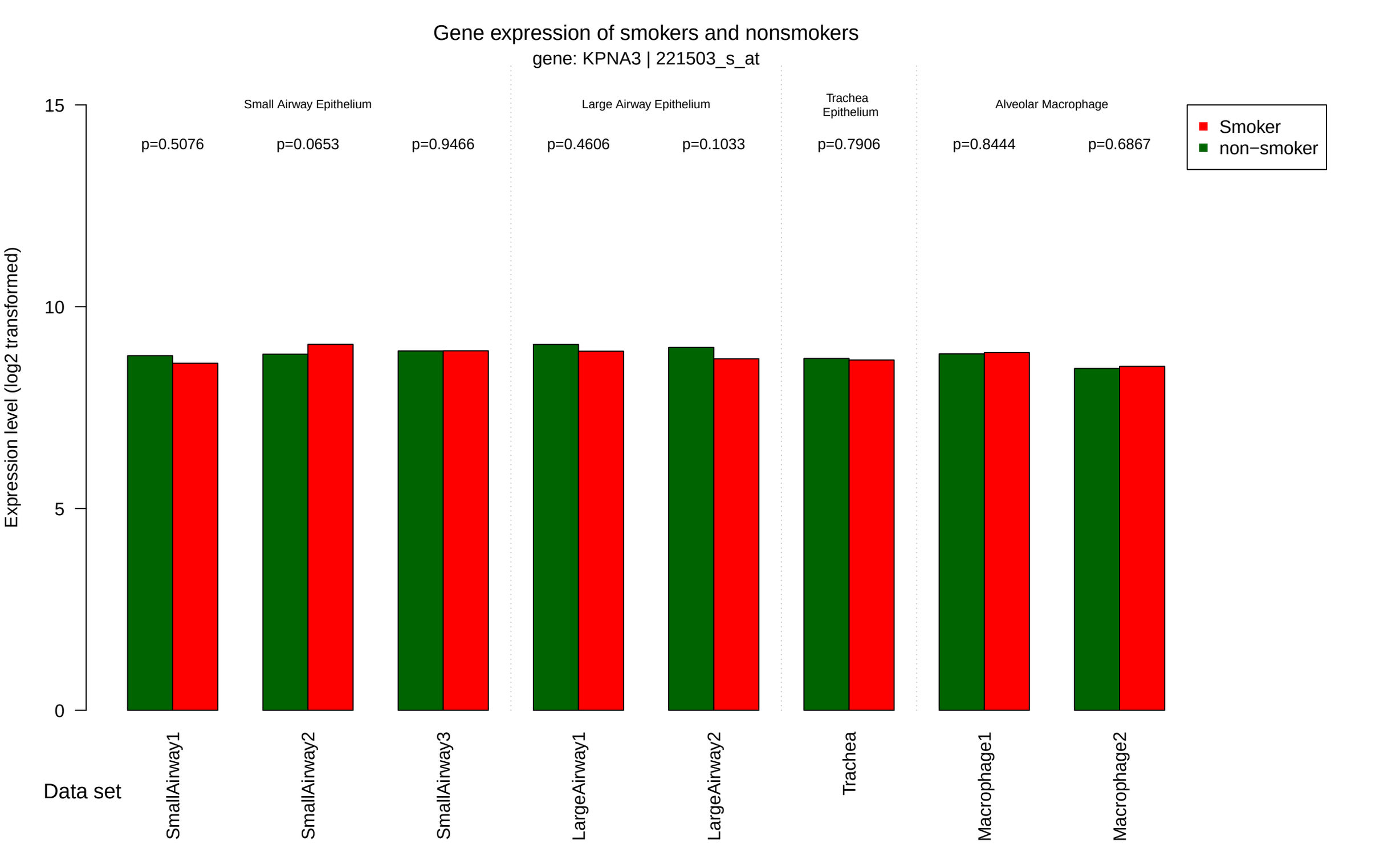
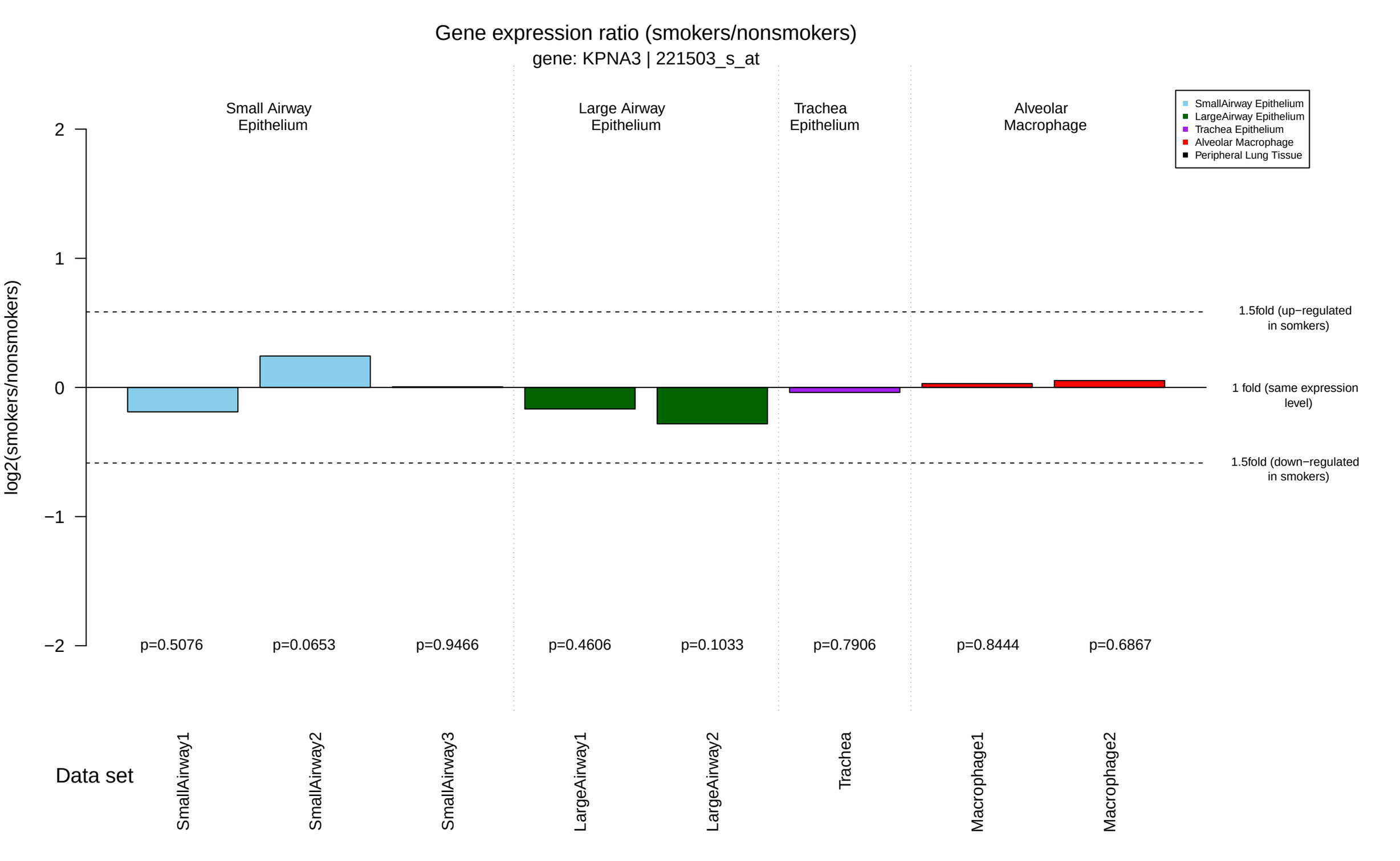
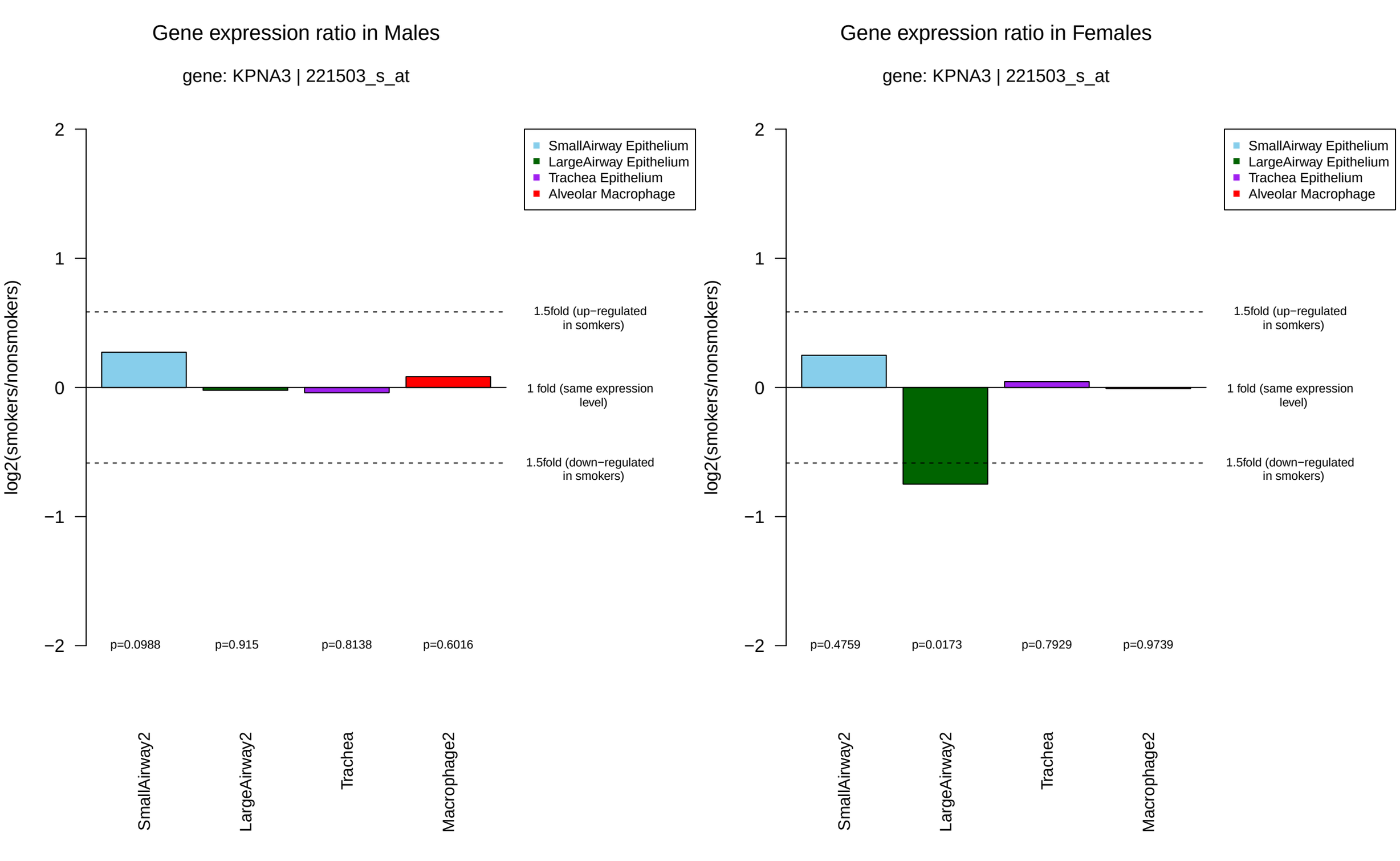
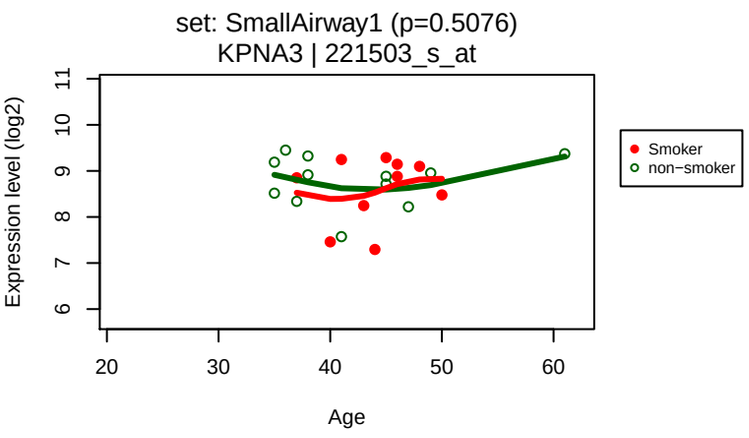
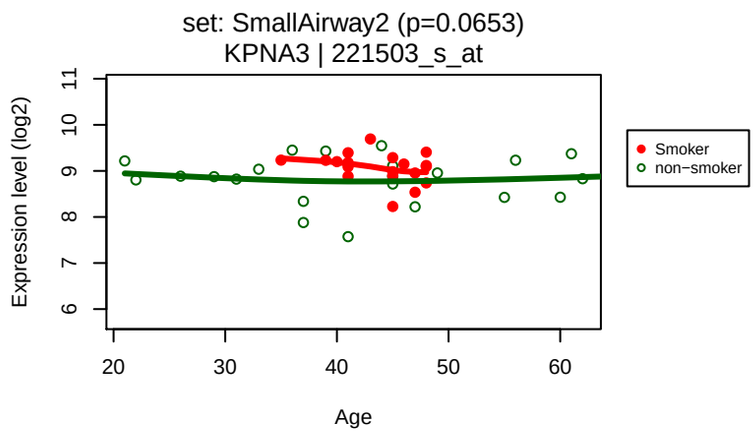
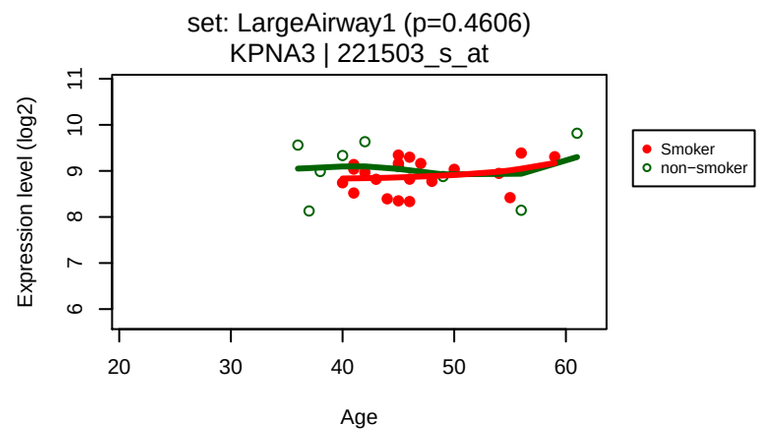

Gene expression of smokers and nonsmokers
gene: KPNA3 | 221503_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5076
p=0.0653
p=0.9466
p=0.4606
p=0.1033
p=0.7906
p=0.8444
p=0.6867
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

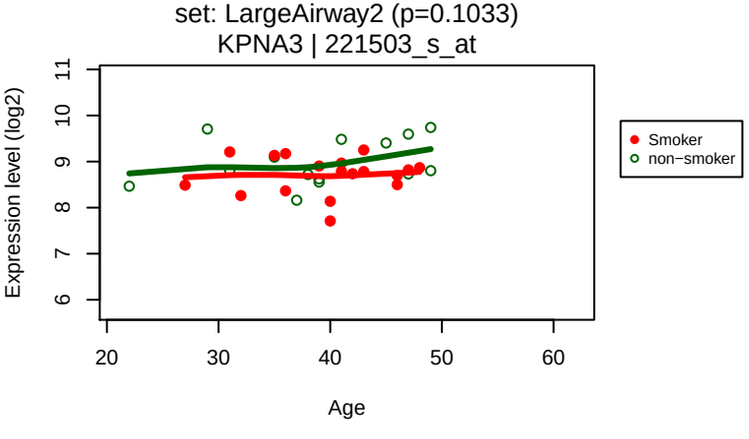

Gene expression ratio (smokers/nonsmokers)
gene: KPNA3 | 221503_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5076
p=0.0653
p=0.9466
p=0.4606
p=0.1033
p=0.7906
p=0.8444
p=0.6867
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

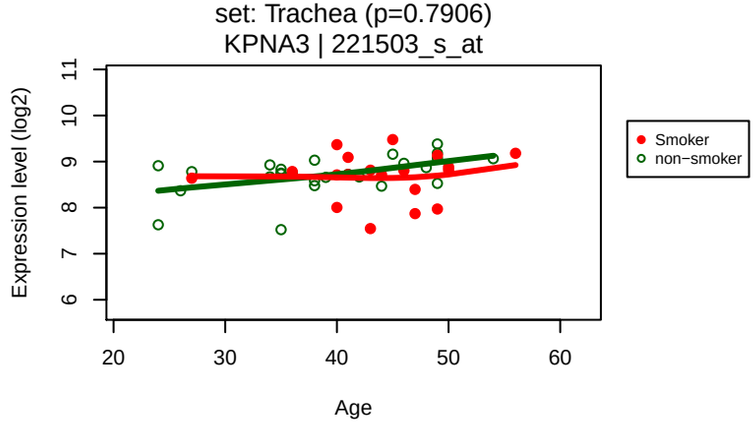

Gene expression ratio in Males
gene: KPNA3 | 221503_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0988
p=0.915
p=0.8138
p=0.6016
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA3 | 221503_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4759
p=0.0173
p=0.7929
p=0.9739
SmallAirway2
LargeAirway2
Trachea
Macrophage2

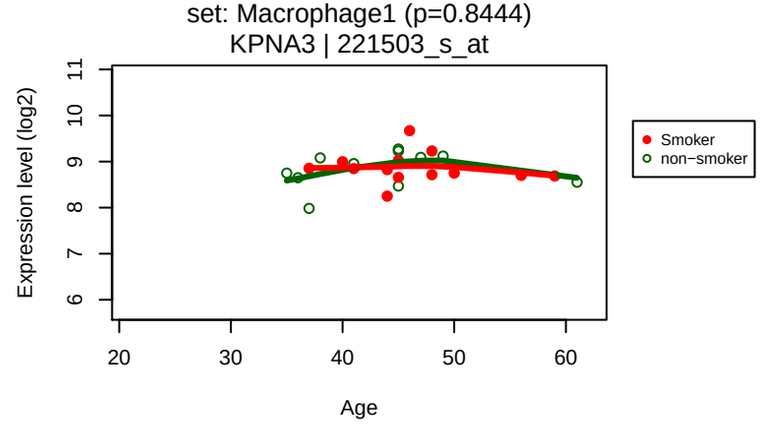

set: SmallAirway1 (p=0.5076)
KPNA3 | 221503_s_at
Expression level (log2)
Age
Smoker
non−smoker

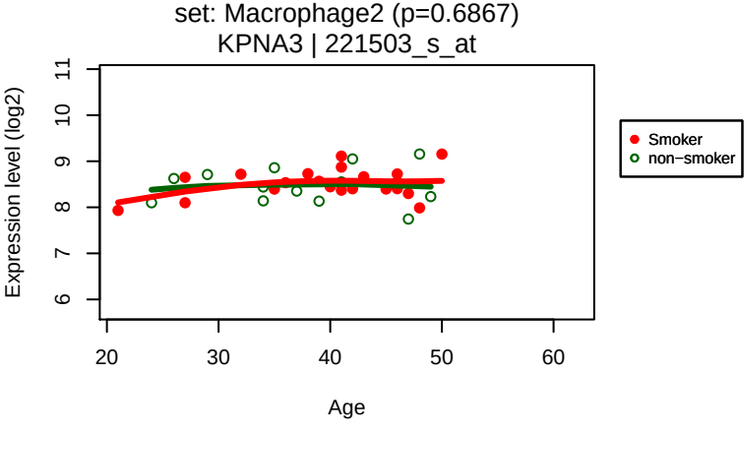

set: SmallAirway2 (p=0.0653)
KPNA3 | 221503_s_at
Expression level (log2)
Age
Smoker
non−smoker

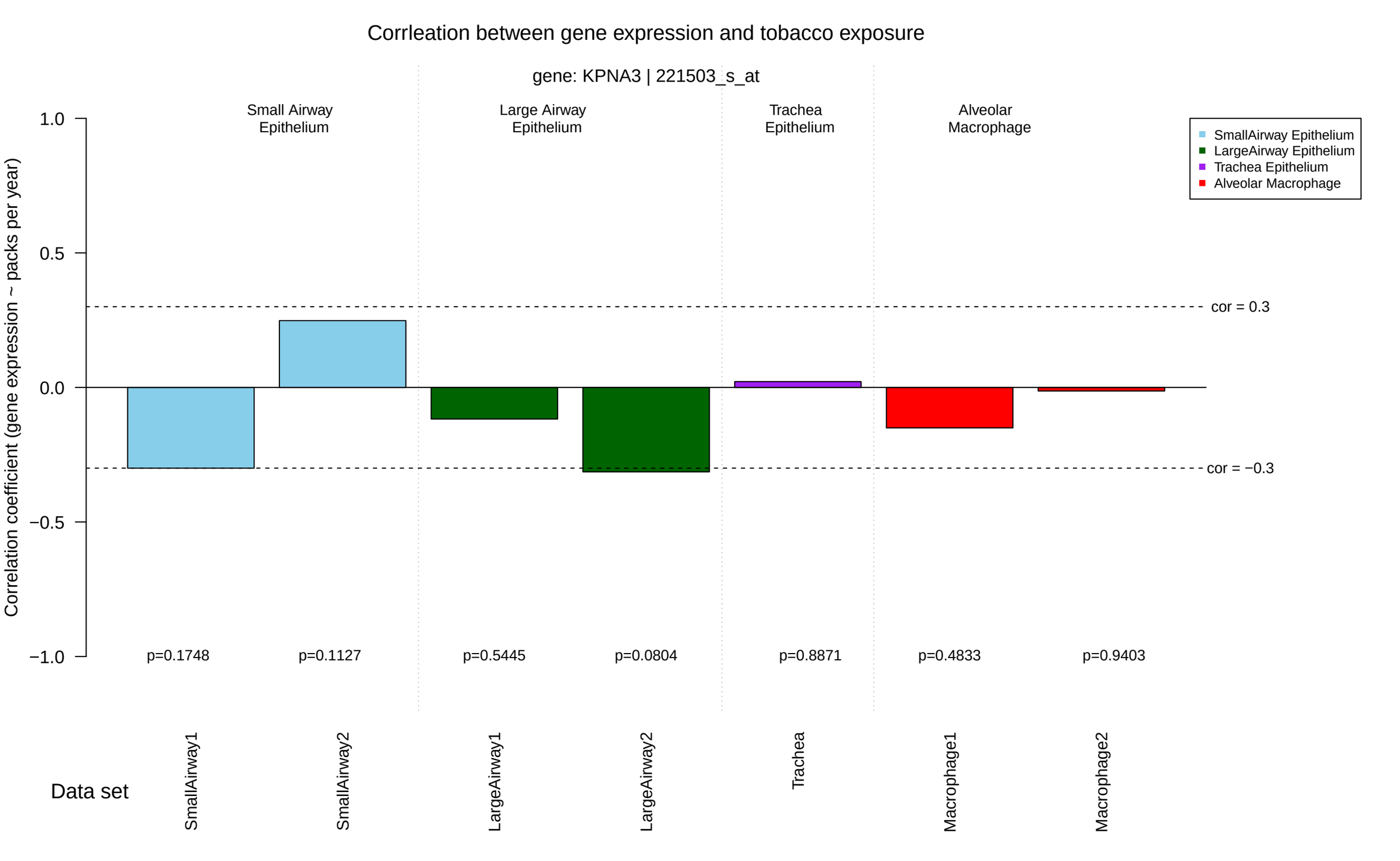

set: LargeAirway1 (p=0.4606)
KPNA3 | 221503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1033)
KPNA3 | 221503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7906)
KPNA3 | 221503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8444)
KPNA3 | 221503_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6867)
KPNA3 | 221503_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA3 | 221503_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1748
p=0.1127
p=0.5445
p=0.0804
p=0.8871
p=0.4833
p=0.9403
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2